Distributed Systems Course | UM6P

Gossip Protocol

in Peer-to-Peer Networks

Task Division, Dependencies & Project Timeline

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March 2026 — 1-Month Project Schedule

https://github.com/MTheCreator/Gossip-Protocol-in-P2P-Systs

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1. Project Overview

This document outlines the **task assignment**, **technical design decisions**, **component dependencies**, and the **1-month project timeline** for the Gossip Protocol implementation. The protocol is decomposed into four components, each assigned to one team member to enable fully parallel development.

Repository: `https://github.com/MTheCreator/Gossip-Protocol-in-P2P-Systs`

1.1. Agreed Design Choices

#	Component	Chosen Strategy
C1	**Peer Discovery**	Kademlia DHT — XOR distance, k-bucket routing, iterative `FIND_NODE`, single bootstrap node
C2	**Message Modeling**	Total ordering via **Lamport timestamps**; extensible struct: `content`, `sender_id`, `timestamp`, `counter`, `channel`
C3	**Dissemination & Failure**	Probabilistic forwarding + receipt counters + **heartbeat-based** failure detection with exponential back-off
C4	**Edge Probability Updates**	**Usage-based reinforcement** — increase on success, slow decay on inactivity

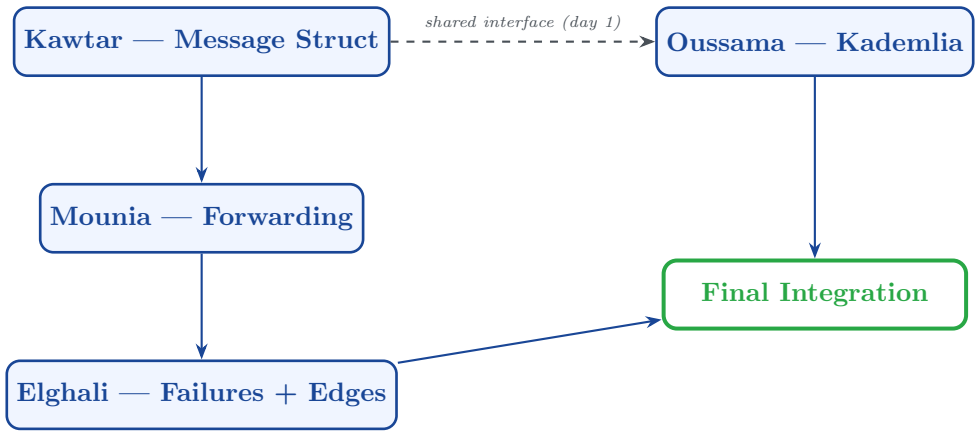
2. Task Division

Each team member owns one component end-to-end: design, implementation, unit testing, and documentation.

Person	Component	Detailed Responsibilities
Oussama Laaroussi	**C1 — Peer Discovery**	XOR distance metric and k-bucket routing table. Iterative `FIND_NODE` lookup (converges in $O(\log n)$ hops). Bootstrap node contact at join time. Enforce $f_{\min}/f_{\max}$ neighbor bounds. Trigger re-discovery whenever a neighbor is lost.
Kawtar Labzae	**C2 — Message Modeling**	Define and **freeze the shared message struct on Day 1** (unblocks the whole team). Implement Lamport clock: increment on send, `max+1` on receive. Buffer out-of-order messages for total-order delivery. Design extensible format for optional metadata fields.
Mounia Baddou	**C3a — Gossip Dissemination**	Forwarding engine: sample each edge with its probability p_e, simulate node drop with configurable probability. Receipt counter per message — once threshold exceeded, permanently ignore. Duplicate detection and logging. Depends on Kawtar's message struct.
Elghali Benjelloun	**C3b + C4 — Failures & Edge Updates**	Heartbeat module: send every Δt, mark suspect at $3\Delta t$, declare dead at $6\Delta t$. Exponential back-off timer on edge failure; reconnect hook back to C1. Usage-based edge probability: reinforce on successful forward, apply slow decay otherwise. Integrate edge feedback with Mounia's forwarding engine.

3. Dependencies & Integration Order

3.1. Dependency Diagram



4. Project Timeline — 1 Month

The project runs from **March 1** to **March 31, 2026**, divided into 4 weeks across four phases.

4.1. Phase Summary

Week	Phase	Goal
1	**Foundation**	Shared interfaces, repo setup, component scaffolding
2	**Core Build**	Full implementation of all four components in parallel
3	**Integration**	Connect components, simulation engine, end-to-end tests
4	**Finalization**	Report, presentation, deployment docs, final polish

4.2. Detailed Weekly Breakdown

Week	Dates	Tasks	Owner(s)
1	**Mar 1 – 7**	**[All]** Kick-off meeting, clone repo, agree on language/framework and branch strategy. **[Kawtar]** Define & freeze `Message` struct + Lamport clock API by **end of day 2**. Push to `main`. **[Oussama]** Start Kademlia: XOR distance metric, k-bucket skeleton. **[Mounia]** Scaffold forwarding engine once struct is available. **[Elghali]** Scaffold heartbeat sender/receiver module.	All
2	**Mar 8 – 14**	**[Oussama]** Complete `FIND_NODE` iterative lookup, bootstrap contact, $f_{\min}/f_{\max}$ enforcement, re-discovery trigger. Unit tests. **[Kawtar]** Finish Lamport ordering, out-of-order buffering. Unit tests. **[Mounia]** Per-edge probabilistic forwarding, message drop simulation, receipt counter, duplicate detection. Unit tests. **[Elghali]** Suspect/dead timeouts ($3\Delta t$ / $6\Delta t$), back-off timer, usage-based edge probability update. Unit tests.	All
3	**Mar 15 – 21**	**[Mounia + Elghali]** Integrate edge-feedback loop (probability update hooks into forwarding calls). **[Oussama + Kawtar]** Connect C1 re-discovery trigger to C3 failure events. **[All]** Full system integration, build simulation engine with configurable topology, first end-to-end smoke test, visualisation of message propagation and edge probability graphs.	All
4	**Mar 22 – 31**	**[All]** Strategy comparison: run parameter sweeps, collect metrics. Bug fixing and performance tuning. Write final report and prepare slides. Deployment documentation (real-device config). **Deadline: March 31.**	All

4.3. Visual Gantt Chart

